Fuzzy Sets

Term	Description
Universe of discourse	Set of all information on a given problem. Once the universe is defined, we can define events on the information in the set.

Fuzzy set

A fuzzy set, A, is a set of ordered pairs $(x, \mu_A(x))$ where $x \in X$, X is the universe of discourse of A, and $\mu_A(x) \in [0, 1]$ represents the membership value of x, i.e. a vague measurement of the availability of x in A.

$$A = \{(x_1, \mu_A(x_1)), (x_2, \mu_A(x_2)), ..., (x_n, \mu_A(x_n))\}$$

This idea is unlike classical or crisp sets, where for any value x, we know for sure if it is available in a crisp set A.

According to the Zadeh notation, a fuzzy set is represented as:

$$A = \frac{\mu_A(x_1)}{x_1} + \frac{\mu_A(x_2)}{x_2} + ... + \frac{\mu_A(x_n)}{x_n} = \sum_{i=1}^{n} \frac{\mu_A(x_i)}{x_i}$$

If X is continuous:

$$A(x) = \int_x \frac{\mu_A(x)}{x}$$

Note that $\frac{\mu(x_i)}{x_i}$ represents the ordered pair $\mu(x_i), x_i$, and not division.
Similarly, $\sum$ and $\int$ represent union of ordered pairs, and not actual summation or integration.

Fuzzy set operations

1. Cardinality

$$\text{Cardinality of } A : |A| = \sum_{x \in X} \mu_A(x)$$

2. Relative cardinality

$$\text{Relative cardinality of } A : |A|_{\text{rel}} = \frac{|A|}{|X|}$$

3. Subset

$$A \text{ is a subset of } B : A \subseteq B = \mu_A(x) \leq \mu_B(x) \ \forall x \in X_A \cup X_B$$

4. Compliment

$$\text{Complement of } A : \overline{A} = \sum_{x \in X} \frac{1 - \mu_x}{x}$$

5. Union

$$\text{Union of } A \text{ and } B : A \cup B = \sum_{x \in X_A \cup X_B} \frac{\max(\mu_A(x), \mu_B(x))}{x}$$

6. Intersection

$$\text{Intersection of } A \text{ and } B : A \cup B = \sum_{x \in X_A \cup X_B} \frac{\min(\mu_A(x), \mu_B(x))}{x}$$

Fuzzy set properties

1. Commutativity

$$A \cup B = B \cup A$$
$$A \cap B = B \cap A$$

2. Associativity

$$A \cup (B \cup C) = (A \cup B) \cup C$$
$$A \cap (B \cap C) = (A \cap B) \cap C$$

3. Distributivity

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

4. Idempotency

$$A \cup A = A$$
$$A \cap A = A$$

5. Involution

$$\overline{\overline{A}} = A$$

6. De Morgan's law

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$
$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

7. Law of absorption

$$A \cup (A \cap B) = A$$
$$A \cap (A \cup B) = A$$